

Configuration Guide

Devices Used

- 1 Cisco 2911 Router (ISP)
- 1 ASA 5506-X Firewall
- 1 Cisco 3560 Layer 3 Switch
- 4 Cisco 2960 Switches
- 1 Server
- 1 Wireless Router
- 11 PCs
- 1 Laptop
- 1 Smartphone

Enterprise IP Addressing Plan

VLAN	Department	Network	Gateway
10	Management	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1
30	Finance	192.168.30.0/24	192.168.30.1
40	IT	192.168.40.0/24	192.168.40.1
50	Wireless	192.168.50.0/24	192.168.50.1
99	Servers	192.168.99.0/24	192.168.99.1

WAN Addressing

Device	Interface	IP
ISP Router	G0/0	100.100.100.1/30
ASA Outside	G1/1	100.100.100.2/30
ASA Inside	G1/2	172.16.1.1/30
Layer3 Switch	G0/1	172.16.1.2/30

STEP 1 – Configure ISP Router

```
enable
configure terminal
hostname ISP
interface GigabitEthernet0/0
description TO_CLOUD
ip address 200.1.1.1 255.255.255.252
no shutdown
exit

interface GigabitEthernet0/1
description TO_FIREWALL
ip address 100.100.100.1 255.255.255.252
no shutdown
exit

end
write
```

STEP 2 – Configure ASA Firewall

```
enable
configure terminal
hostname ASA_FIREWALL
```

Outside Interface

```
interface GigabitEthernet1/1
nameif outside
security-level 0
ip address 100.100.100.2 255.255.255.252
no shutdown
exit
```

Inside Interface

```
interface GigabitEthernet1/2
nameif inside
security-level 100
ip address 172.16.1.1 255.255.255.252
no shutdown
exit
```

Default Route

```
route outside 0.0.0.0 0.0.0.0 100.100.100.1
```

write memory

STEP 3 – Configure Layer 3 Switch

enable

configure terminal

hostname CORE_SWITCH

ip routing

Create VLANs

vlan 10

name MANAGEMENT

vlan 20

name HR

vlan 30

name FINANCE

vlan 40

name IT

vlan 50

name WIRELESS

vlan 99

name SERVERS

Configure Routed Interface to ASA

interface GigabitEthernet0/1

no switchport

ip address 172.16.1.2 255.255.255.252

no shutdown

Configure the DHCP Server Port

Since your DHCP server is connected to **GigabitEthernet0/2**, place it in VLAN 99.

interface GigabitEthernet0/2

switchport mode access

switchport access vlan 99

no shutdown

Configure SVIs (Gateways)

```
interface vlan10
ip address 192.168.10.1 255.255.255.0
ip helper-address 192.168.99.10
no shutdown
```

```
interface vlan20
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.99.10
no shutdown
```

```
interface vlan30
ip address 192.168.30.1 255.255.255.0
ip helper-address 192.168.99.10
no shutdown
```

```
interface vlan40
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.99.10
no shutdown
```

```
interface vlan50
ip address 192.168.50.1 255.255.255.0
ip helper-address 192.168.99.10
no shutdown
```

```
interface vlan99
ip address 192.168.99.1 255.255.255.0
no shutdown
```

Configure Access Switch Connections

Since each Layer 2 switch serves only one VLAN, configure the uplinks as **access ports** rather than trunks.

Management Switch

```
interface FastEthernet0/1
switchport mode access
switchport access vlan 10
no shutdown
```

HR Switch

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 20
no shutdown
```

Finance Switch

```
interface FastEthernet0/3
switchport mode access
switchport access vlan 30
no shutdown
```

IT Switch

```
interface FastEthernet0/4
switchport mode access
switchport access vlan 40
no shutdown
```

Wireless Router

```
interface FastEthernet0/5
switchport mode access
switchport access vlan 50
no shutdown

interface vlan50

ip address 192.168.50.1 255.255.255.0

ip helper-address 192.168.99.10

no shutdown
```

Default Route

```
ip route 0.0.0.0 0.0.0.0 172.16.1.1

end

write
```

STEP 4 – Configure the Management Switch

```
enable
configure terminal
hostname MGMT_SW
interface FastEthernet0/1
switchport mode access
switchport access vlan 10
```

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 10
```

```
interface FastEthernet0/3
switchport mode access
switchport access vlan 10
```

```
end
write
```

- Fa0/1 is the uplink to the Layer 3 switch.
- Fa0/2 connects to PC1.
- Fa0/3 connects to PC2.

Configure the HR Switch

```
enable
configure terminal
hostname HR_SW
```

```
interface FastEthernet0/1
switchport mode access
switchport access vlan 20
```

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 20
```

```
interface FastEthernet0/3
switchport mode access
switchport access vlan 20
```

```
end
write
```

Configure the Finance Switch

```
enable
configure terminal
hostname FIN_SW

interface FastEthernet0/1
switchport mode access
switchport access vlan 30

interface FastEthernet0/2
switchport mode access
switchport access vlan 30

interface FastEthernet0/3
switchport mode access
switchport access vlan 30

end
write
```

Configure the IT Switch

```
enable
configure terminal
hostname IT_SW

interface FastEthernet0/1
switchport mode access
switchport access vlan 40

interface FastEthernet0/2
switchport mode access
switchport access vlan 40

interface FastEthernet0/3
switchport mode access
switchport access vlan 40

end
write
```

STEP 5 – Configure the DHCP Server

Desktop → IP Configuration

- IP Address: 192.168.99.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.99.1
- DNS Server: 8.8.8.8

Services → DHCP → On

Create these pools:

Pool	Default Gateway	Start IP
Management	192.168.10.1	192.168.10.100
HR	192.168.20.1	192.168.20.100
Finance	192.168.30.1	192.168.30.100
IT	192.168.40.1	192.168.40.100
Wireless	192.168.50.1	192.168.50.100

Use subnet mask 255.255.255.0 for each pool.

STEP 6 – Configure the Wireless Router

If connected via a LAN port:

- Disable DHCP.
- Set LAN IP to 192.168.50.2/24.
- Default Gateway: 192.168.50.1.
- SSID: EnterpriseWiFi.
- Security: WPA2-PSK.

STEP 7 – Configure End Devices

For every PC:

- Desktop → IP Configuration → **DHCP**.

For the Laptop and Smartphone:

- Connect to the EnterpriseWiFi SSID.
- Enter the WPA2 password.
- Obtain an IP address automatically (DHCP).

Layer 2 Switch Management IP

This is something many beginners miss.

Even though Layer 2 switches don't perform routing, they **can have one management IP address** so you can ping them or manage them remotely.

Management Switch

```
enable
configure terminal
hostname MGMT_SW
interface vlan 10
ip address 192.168.10.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.10.1
end
write
```

HR Switch

```
enable
configure terminal
hostname HR_SW
interface vlan 20
ip address 192.168.20.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.20.1
end
write
```

Finance Switch

```
enable
configure terminal
hostname FIN_SW
interface vlan 30
ip address 192.168.30.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.30.1
end
write
```

IT Switch

```
enable
configure terminal
hostname IT_SW
interface vlan 40
ip address 192.168.40.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.40.1
end
write
```

Why Assign Management IPs?

Now you can ping the switches from any PC.

For example, from a Management PC:

```
ping 192.168.10.2
```

From an IT PC:

```
ping 192.168.20.2
```

Because the Layer 3 switch performs inter-VLAN routing, all of these management IPs are reachable (provided your routing and firewall rules allow it).

Network Address Summary

Device	IP Address
Core L3 Switch VLAN10	192.168.10.1
Management Switch	192.168.10.2
Core L3 Switch VLAN20	192.168.20.1
HR Switch	192.168.20.2
Core L3 Switch VLAN30	192.168.30.1
Finance Switch	192.168.30.2
Core L3 Switch VLAN40	192.168.40.1
IT Switch	192.168.40.2
Core L3 Switch VLAN50	192.168.50.1

Device	IP Address
Wireless Router	192.168.50.2
Core L3 Switch VLAN99	192.168.99.1
DHCP Server	192.168.99.10

STEP 1: Configure Static Routes

ISP Router

The ISP Router only knows the directly connected networks. You must tell it how to reach all your internal VLANs.

```
enable
configure terminal

ip route 192.168.10.0 255.255.255.0 100.100.100.2
ip route 192.168.20.0 255.255.255.0 100.100.100.2
ip route 192.168.30.0 255.255.255.0 100.100.100.2
ip route 192.168.40.0 255.255.255.0 100.100.100.2
ip route 192.168.50.0 255.255.255.0 100.100.100.2
ip route 192.168.99.0 255.255.255.0 100.100.100.2

end
write
```

ASA Firewall

You already configured

```
route outside 0.0.0.0 0.0.0.0 100.100.100.1
```

Now add a route to the internal networks.

```
enable
configure terminal
route inside 192.168.0.0 255.255.0.0 172.16.1.2
write memory
```

This tells the firewall that all internal networks are reachable via the Layer 3 switch.

Layer 3 Switch

You already configured

```
ip route 0.0.0.0 0.0.0.0 172.16.1.1
```

Nothing more is required.

STEP 2: Configure NAT on ASA Firewall

Without NAT, internal PCs cannot access the ISP network.

Enter configuration mode.

```
enable
configure terminal
```

Define the inside network.

```
object network INTERNAL_NETWORK
```

```
subnet 192.168.0.0 255.255.0.0
```

Enable PAT (Port Address Translation).

```
nat (inside,outside) dynamic interface
```

```
write memory
```

Now all VLANs will use the outside interface IP (100.100.100.2) when communicating with the ISP.

STEP 3: Configure Firewall ACLs

Your external devices are

EX1

EX2

EX3

Let's assume

Device IP

EX1 201.1.1.10

EX2 201.1.1.20

EX3 201.1.1.30

Block these addresses.

```
access-list OUTSIDE-IN extended deny ip host 201.1.1.10 any
access-list OUTSIDE-IN extended deny ip host 201.1.1.20 any
access-list OUTSIDE-IN extended deny ip host 201.1.1.30 any
access-list OUTSIDE-IN extended permit ip any any
```

Apply ACL

access-group OUTSIDE-IN in interface outside
write memory

STEP 4: Configure the Cloud

Packet Tracer's Cloud device is just a media simulator and doesn't require CLI configuration. Connect it as follows:

Cloud Gi4
|
ISP Router G0/0

Assign the ISP Router interface:

interface GigabitEthernet0/0

ip address 200.1.1.1 255.255.255.252

no shutdown

The cloud itself remains unconfigured.

STEP 5: Configure External Devices

Assign static IPs.

EX1	EX2	EX3
IP Address 201.1.1.10	IP Address 201.1.1.20	IP Address 201.1.1.30
Subnet Mask 255.255.255.0	Subnet Mask 255.255.255.0	Subnet Mask 255.255.255.0
Gateway 200.1.1.1	Gateway 200.1.1.1	Gateway 200.1.1.1
DNS 8.8.8.8	DNS 8.8.8.8	DNS 8.8.8.8

STEP 6: Configure PCs

For every wired PC

Desktop

↓

IP Configuration

↓

DHCP

The DHCP server should assign addresses such as:

Management

192.168.10.100

Gateway

192.168.10.1

DNS

8.8.8.8

HR

192.168.20.100

Finance

192.168.30.100

IT

192.168.40.100

Wireless

192.168.50.100

STEP 7: Verify DHCP

On every PC

Desktop

Command Prompt

Type

ipconfig

Example

IP Address

192.168.10.100

Subnet

255.255.255.0

Gateway

192.168.10.1

DNS

8.8.8.8

If all departments receive the correct network address, DHCP is working correctly.

STEP 8: Verify Inter-VLAN Routing

From the Management PC:

ping 192.168.20.100

(HR PC)

ping 192.168.30.100

(Finance PC)

ping 192.168.40.100

(IT PC)

ping 192.168.50.100

(Laptop)

All replies should be successful.

STEP 9: Test the Firewall

From **EX1**, ping the Management PC: ping 192.168.10.100

Expected result:

Request timed out.

Repeat from EX2 and EX3. They should also be blocked by the ASA ACL.

STEP 10: Test Internet Access

From the Management PC:

```
ping 100.100.100.1
```

(ASA outside next hop)

Expected:

```
Reply from 100.100.100.1
```

Then ping the ISP router's cloud-facing interface:

```
ping 200.1.1.1
```

If NAT and routing are configured correctly, this should also succeed.

Final Verification Checklist

Feature	Status
VLANs Configured	✓
Layer 3 Inter-VLAN Routing	✓
DHCP Server Assigning IPs	✓
DHCP Relay (ip helper-address)	✓
Wireless Clients Receiving IPs	✓
Static Routing	✓
Default Routes	✓
ASA Firewall	✓
PAT (NAT Overload)	✓
ACL Blocking EX1, EX2, EX3	✓
Internal Departments Can Ping Each Other	✓
Internal Devices Can Reach ISP	✓
External Devices Blocked from Internal Network	✓